

Injection Measurements and Simulation for a Floating Gate MOSFET Designed for Radiation Measurements

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Abstract—In this paper, a Floating Gate transistor designed and fabricated to be used as a radiation sensor is presented, and the charge injection process through an injection electrode is studied. The device was successfully charged, and the injection process was simulated in a SPICE software, considering that the injection mechanism was Fowler-Nordheim tunneling. The parameters for the Fowler-Nordheim current were found and simulations successfully reproduced the measurements. These simulations will allow to improve the desing and the charge injection setup.

Index Terms—Floating-Gate transistor; Fowler-Nordheim; Dosimetry; SPICE Simulation

I. INTRODUCTION

According to the International Commission for Radio-Protection (ICRP), in-vivo dosimetry is recommended for safety and quality assurance (QA) in radiotherapy applications [1], [2]. In-vivo dosimetry is an important practice not only to prevent accidents, but also to ensure that the planned treatment is effectively carried out. Recommendations suggest that the delivered dose should not vary from the planned treatment more than 3% en each session. In this context, it is necessary for the dosimeters to be highly accurate.

For the last decades, MOS dosimeters have been proposed and used for medical applications as they present a series of advantages [3], [4]. Nevertheless, there are still a series of issues that must be addressed. In the last years, in order to overcome some of these issues, we have proposed the use of Field Oxide as a thick gate oxide (FOXFET), proving an improvement in the dosimeter sensitivity [5]. The FOXFET has shown to be appropriate for radiotherapy [6], even though

the thick gate oxide also exhibited some instability in the trapped charge, manifested as fading. This dosimeter was also used to implement several differential measuring methods [7]–[9], but the differential method implies the use of different gate voltages applied each transistor, hence an electric connection with the dosimeter is needed, which is undesired in medical applications.

In the MOS devices family, the floating gate (FG) MOSFET is an alternative device to be used as a dosimeter, as shown in [10]–[14]. According to [11], [12], the FGMOSFET sensitivity can be improved extending the floating gate over the field oxide, in a similar way as the FOXFET, but also it presents the advantage that if the FG is charged, the electric field within the oxide can be maintained without external power supply [12], getting rid of the electrical connection during irradiation.

In order to achieve an optimal performance of the dosimeter, it is important to successfully dominate the floating gate transistor as a device, which means primarily, to have a full understanding of the charge injection process. In this paper, a FG device without control gate and with injection electrode is presented. The physical design considering its future application to radiation dosimetry is shown as well as first measurements regarding the injection process and its electrical simulation using SPICE software. A successful simulation of this process will allow to enhance the dosimeter physical design and also to design proper peripheral circuitry integrated with the dosimeter for automatic on chip charge injection.

II. FLOATING GATE DESIGN

The Floating Gate (FG) dosimeter was designed and fabricated in a CMOS 0.5 μm process. An schematic diagram of the

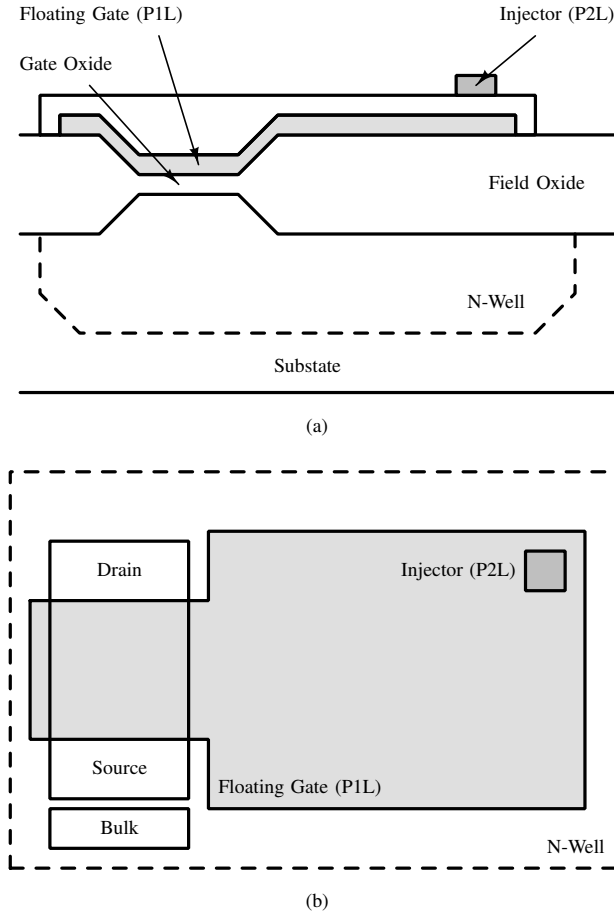


Fig. 1. Simplified diagram for the Floating Gate physical design. (a) The cross-section shows how the different polysilicon layers overlap, and gives an idea of the different oxides involved in the device structure. (b) A simplified version of the layout to understand the different areas involved. For better understanding, metal, contacts and n/p diffusion layers were omitted. Bulk and Source are shorted from fabrication.

device is shown in figure 1. The device is a p-channel MOSFET, where the polysilicon layer (P1L) acts as the floating gate itself and is extended over the Field Oxide to improve radiation sensitivity [5], [11], [12]. Both the gate oxide (GOX) and the field oxide (FOX) provide radiation generated charge [17] that will be collected in the floating gate (Q_{FG}), giving place to a shift in the floating gate potential (V_{FG}). To maximize the sensitivity of V_{FG} with radiation dose, the P1L over FOX should be much larger than the P1L over GOX, in order to counter the difference in oxide thickness, which are $t_{FOX} \simeq 288\text{ nm}$ and $t_{GOX} \simeq 14\text{ nm}$, respectively. A ratio $A_{FOX}/A_{GOX} = 100$ was chosen, and considering the width and length of the active device to be $W = L = 20\text{ }\mu\text{m}$, the area for each P1L are $A_{fox} = 400\text{ }\mu\text{m} \times 100\text{ }\mu\text{m}$ and $A_{gox} = 20\text{ }\mu\text{m} \times 20\text{ }\mu\text{m}$.

An important aspect of the design process was to determine the layout for the injection of charge in the floating gate. Charging the floating gate occurs by Fowler-Nordheim (FN) tunneling from the injector electrode to the floating gate [16]. When a voltage is applied to the injector (V_{inj}), the floating

gate (V_{FG}) potential due to the capacitive coupling of the conducting polysilicon strips is

$$V_{FG} = \frac{C_{inj}V_{inj} + (C_{fox} + C_{gox})V_{FET} + Q_{FG}}{C_{inj} + C_{fox} + C_{gox}}, \quad (1)$$

where C_{inj} is the capacitance between the injector and the floating gate, C_{fox} and C_{gox} are the capacitance of the field oxide and gate oxide respectively, V_{FET} is the voltage commonly applied to the drain, source, and bulk of the FGMOSFET, and Q_{FG} is the charge accumulated in the Floating Gate. For injection to occur, V_{FG} must be different to V_{inj} , hence the voltage applied to the injector should drop mainly across the interpoly oxide and not in the oxide below the FG, which is P1L. Considering $Q_{FG} \simeq 0$ in equation 1, this happens when $(C_{fox} + C_{gox}) \gg C_{inj}$.

The injection electrode was implemented with a second polysilicon layer (P2L) available, which lies above the P1L. Taking into account the capacitance ratio, the P2L area was designed as little as possible, considering the metal contact and the fabrication process specifications, resulting in $A_{inj} = 6\text{ }\mu\text{m} \times 6\text{ }\mu\text{m}$. No Control Gate was included over the Floating Gate as the parasitic capacitance of the control gate impoverishes radiation sensitivity [11].

III. EXPERIMENTAL MEASUREMENTS OF THE INJECTION PROCESS

The Floating Gate was designed and fabricated. As this is the first characterization of the device, the fabricated dies were not wire-bonded nor encapsulated. The electrical characterization was performed in a probe station.

Figure 2 shows the schematic circuits for both the current-voltage (IV) measurement for the FG MOSFET (fig. 2.a) and the charge injection process in the floating gate (fig. 2.b). For the IV measurement, two Keithley 617 electrometers were used, one as a voltage source (V_{SD}) and the other one as an ammeter (I_D). The voltage source was swept while measuring the drain current (I_D). The Initial charge condition in the FG was different for each sample. Some of them were discharged, and no current flow since no channel was formed beneath the gate of the device, and another batch was already charged from fab, so a typical transistor output characteristic (I_D vs V_{SD}) was measured.

A key aspect of the FGMOSFET is that the FG itself can be charged or discharged in order to change its operation state. As the samples are p-channel MOSFETs, the FG should be charged with electrons to induce a conduction channel underneath the interface. For charge injection, a Keithley 220 programmable current source was connected to the injection electrode while a constant $V_{SD} = 100\text{ mV}$ was applied to the MOSFET, and the I_D was monitored. A buffered output from the current source allowed to monitor the injection voltage (V_{inj}) with one of the Keithley 617 as voltmeter.

In order to charge the samples with electrons, a negative current (I_{inj}) was applied, whereas when discharging the samples a positive I_{inj} was applied instead. In both cases, the injection process was similar. V_{inj} increases in absolute value as the

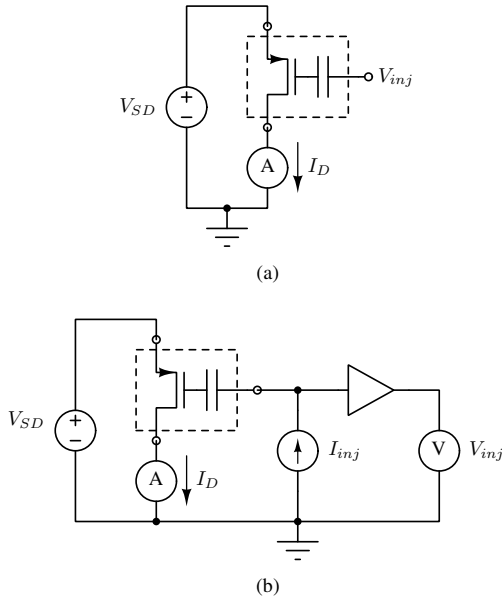


Fig. 2. Schematic circuits for (a) the current-voltage measurement for the FG MOSFET, and (b) the charge injection process.

injection node charges in time until it reached approximately $V_{inj} = -22$ V when charging and $V_{inj} = +18$ V when discharging. At these voltages, the injection starts, and the drain current will change in time.

A. Injected charge determination

The change in the Floating Gate drain current after the injection process was not as expected if a -55 pA current was effectively injected. Considering the standard MOSFET current-voltage characteristics shown in fig. 3 as reference of what is happening in the floating gate, the change in the gate-source voltage (ΔV_{GS}) needed to produce the current variation can be determined. Afterwards, knowing ΔV_{GS} , the amount of charge injected (ΔQ) and the average injection current during the experiment can also be estimated as

$$\Delta Q = (C_{gox} + C_{fox}) \times \Delta V_{GS}, \quad (2)$$

$$\bar{I}_{inj} = \frac{\Delta Q}{t_{inj}}, \quad (3)$$

where t_{inj} is the time of the experiment where the injection is appreciable.

All the experiments carried out lead to calculated average injection current around tens of fA, much lower than the current set in the current source. Nevertheless, these values are consistent with the Fowler-Nordheim expression for the injection current density with the electrical field applied to the oxide [16],

$$I_{inj}(V_{ox}) = A_{inj} C \varepsilon_{ox}^2 \exp\left(-\frac{\varepsilon_o}{\varepsilon_{ox}}\right), \quad (4)$$

where C and ε_o are constants in terms of the effective mass and barrier height [16], $\varepsilon_{ox} = \frac{V_{ox}}{t_{ox}}$ is the electric field across de oxide, $V_{ox} = V_{inj} - V_{FG}$ is the voltage across de oxide, and

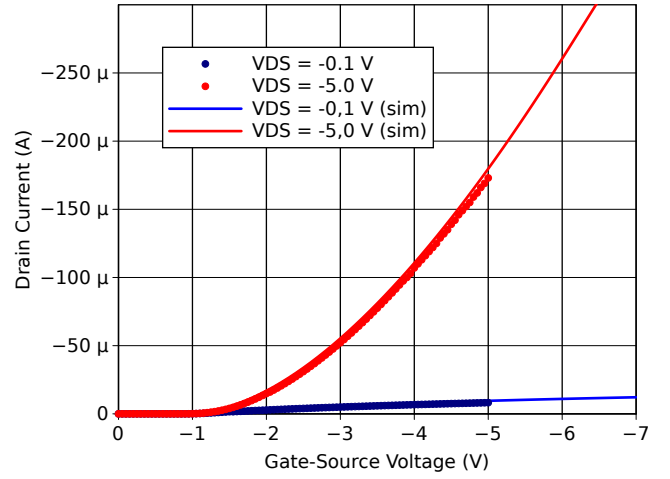


Fig. 3. Transfer characteristics of a standar PMOS transistor used as reference for $V_{DS} = -0.1$ V and $V_{DS} = -5.0$ V, measured and simulated showing good agreement.

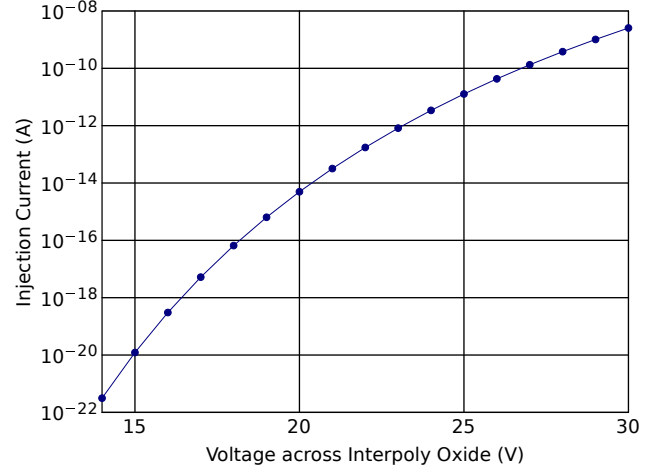


Fig. 4. Fowler-Norheim current through the interpoly oxide as a function of the voltage applied to the structure.

t_{ox} is the interpoly oxide thickness, which is 33 nm thick for this process. Figure 4 shows the injection current dependence with the voltage across the injection electrode, calculated for the geometry of the designed transistor. The figure shows that the theoretical injection current is around what we estimated from charge injection. This result suggests that most of the current is lost in the experimental setup, making it difficult to estimate in advance the outcome of the experiment.

IV. ELECTRICAL SIMULATION OF THE INJECTION PROCESS

To have a better understanding of the injection process, we propose to elaborate an electrical model that can be simulated using SPICE software. The model includes a voltage controlled current source (I_{inj}) that implements equation 4, emulating the charge injection process through the interpoly oxide. This current source model a non-linear resistance which is connected parallel to the interpoly capacitance represented

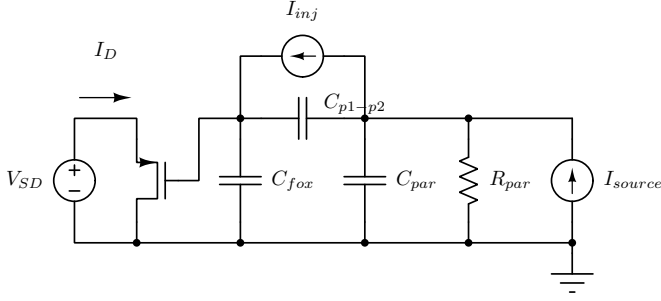


Fig. 5. Circuit schematic for the simulation of the charge injection process.

with $C_{p1-p2} = 32.4$ fF. It also includes the parasitic resistance (R_{par}) and capacitance (C_{par}) from the experimental setup, the p-channel MOSFET with simulation model provided by the foundry, and the field oxide capacitance due to the extension of the floating gate over non-active area ($C_{fox} = 4.8$ pF). The constructive parameters were taken from the process specification, and it was necessary to determine the parasitic elements. An schematic for this model is shown in figure 5.

A. Determination of parameters from measurements

To determine the parasitic elements for our simulation, the behavior before injection is analyzed. If no current is injected in the floating gate, the branch in fig. 5 including the current source I_{inj} can be disregarded, and the circuit simplifies to an RC circuit, with characteristic constant τ_{RC} and steady state voltage $V_{SS} = I_{source} \times R_{par}$. Fitting the data to the characteristic equation of an RC circuit, both $R_{par} = 470$ G Ω and $C_{par} = 172$ pF were determined.

Finally, the parameters for the Fowler-Nordheim injection current that best fitted the change in the drain current due to the injection measurements were determined, resulting in $C = 4.61 \mu\text{A}/\text{V}^2$ and $\epsilon_o = 221$ MV/cm. Initial conditions for the FG voltage and the injection electrode voltage were taken from measurements.

B. Simulation results

A first experimental setup was simulated. In this setup, the source current was $I_{source} = -55$ pA, and was turned on at $t_o = 56$ s and was turned off $T = 566$ s later. The initial conditions for the different nodes were $V_{inj} = -1.19$ V and $V_{FG} = -3.54$ V, the voltage needed for the transistor to have an initial drain current $I_D = 7.0$ μA . Figure 6 shows the comparison between measurement and simulation of the injection process. If we analyze the drain current evolution, it can be seen that at first, before the current source is turned on, the drain current remains constant. When the current source is turned on, the drain current increases due to the capacitive coupling between the injection node and the floating gate until approximately 230 s, where the current is approximately 7.2 μA . Afterwards, the injection of charge in the FG is significant, and the increase in the drain current is greater. After 400 s, the rate in the drain current increase slows down, and finally at $t = 622$ s the current source is turned off and

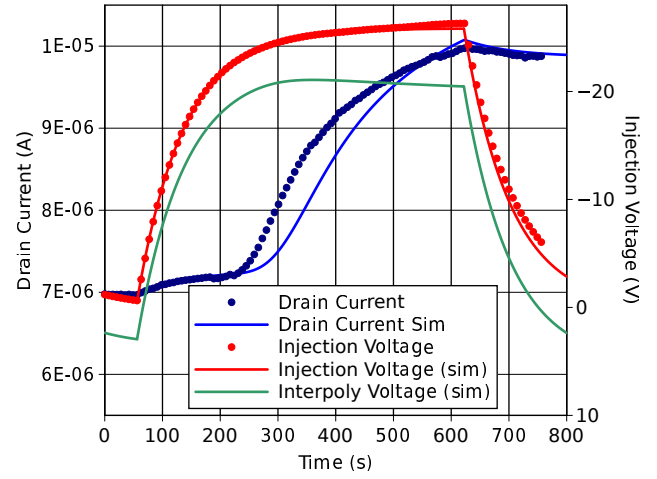


Fig. 6. Injection process for the fabricated device. The injection voltage (red) and the drain current (blue) evolution in time was measured (dots) and simulated (straight line). From simulations, the voltage across the interpoly oxide could be obtained and is shown as a green line.

the drain current decreases once again due to the capacitive coupling between the injector and the FG. The simulation successfully reproduces this behavior.

The advantage of simulating this procedure, is that we can now get a better insight of the injection current behavior. The simulated I_{inj} is plotted in fig. 7 along with the current source current. There is a considerable amount of time where the injection process is negligible, that is since the current source is turned on at $t = 56$ s until approximately $t = 200$ s. Then, the injection current increases considerably until a peak value of 41.3 fA at $t \simeq 350$ s, and then decreases, until the current source is turned off at $t = 566$ s and the injection current quickly falls down to zero. Considering the time where I_{inj} is not negligible, the mean injection current simulated is $\bar{I}_{inj} = 22$ fA. According to our calculations for this experiment, the average injection current considering the same time interval was 23 fA.

Considering that there was good agreement between measurement and simulation for that particular experiment, a second experiment was also simulated, changing only the initial conditions of the experimental setup. This time, the source current was also $I_{source} = -55$ pA, it was turned on at $t_o = 60$ s and was turned off $T = 250$ s later, and the initial condition for the injection node was $V_{inj} = 46$ mV and the FG voltage needed for the transistor to have an initial drain current $I_D = 6.56$ μA was $V_{FG} = -3.335$ V. The FG constructive parameters and the FN parameters were maintained in order to validate the simulation model. Results for this second simulation are shown in fig. 8 and 9. The overall behavior is similar to the previous experiment and simulation: everything is steady until the current source is turned on, then a coupling between the injection voltage and the drain current is observed, and after $t = 200$ s the injection process becomes appreciable. In this experiment, the injection time was shorter, so the injection current does not reach a maximum value during the

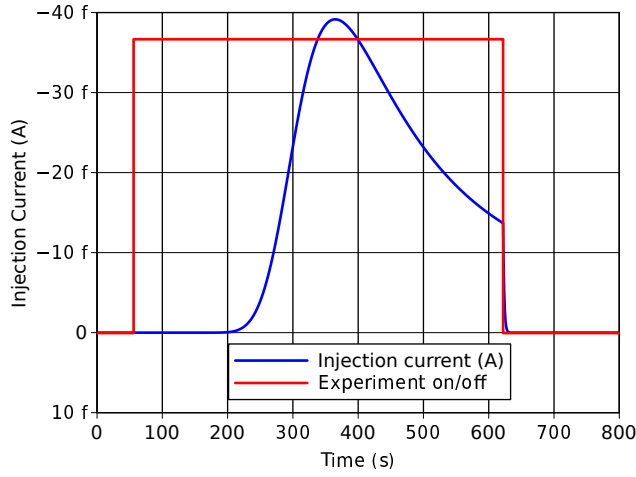


Fig. 7. Injection current taken from simulation in fig. 6. The red line shows when the current source is turned on during the experiment.

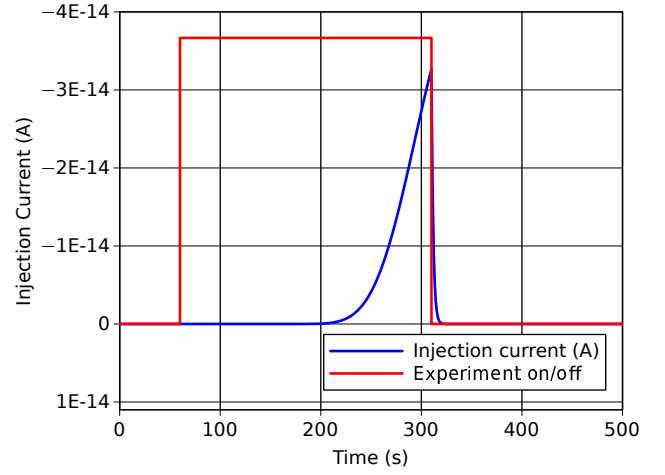


Fig. 9. Injection current taken from simulation in fig. 8. The red line shows when the current source is turned on during the experiment.

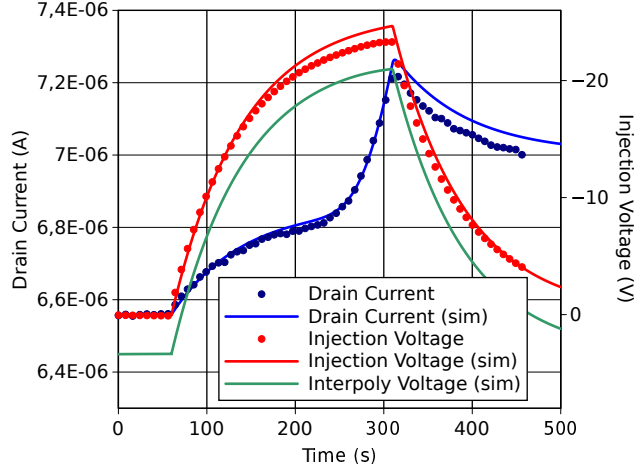


Fig. 8. Injection process under different initial conditions than 6. The simulation showed again good agreement compared to measurements.

injection process, and once again it falls down to zero when the experiment is ended at $t = 310$ s. The mean injection current during this time was 10 fA. According to our calculations for this experiment, the average injection current considering the same time interval was 14 fA, 40% higher.

V. DISCUSSION AND CONCLUSION

In this paper, the design of a floating gate transistor for dosimetry applications is presented. The device was fabricated and first electrical characterizations are presented. Primarily, the charge injection process is studied, which is of most importance to have control of the initial electrical conditions when the device is irradiated and used as a dosimeter.

The injection process was successfully performed, but it was found that due to parasitics in the experimental setup, the amount of charge injected was much less than expected. Taking this into account, it was decided to study this injection process, and simulated in order to have a better understanding

of the limitations. The experimental setup was then modeled with a simple SPICE circuit considering the fabrication process specifications, the parasitics calculated for the setup and the theoretical Fowler-Nordheim injection current. Parameters for the FN current were found to be $C = 4.61 \mu\text{A}/\text{V}^2$ and $\varepsilon_o = 221 \text{ MV}/\text{cm}$, very similar for those reported for injection from a silicon substrate to Polysilicon gate through a SiO_2 oxide [18]. As this is a polysilicon to polysilicon injection, it is of no surprise that these constants are slightly different.

Simulations successfully reproduce the behavior measured during the injection process, including the capacitive coupling between the injection electrode and the floating gate. Moreover, it was possible to observe the behavior of the injection current during the simulated experiment, which was as expected: at the beginning of the experiment the injection current is negligible, then it grows rapidly, increasing the FG electric potential, which eventually lowers the voltage difference and the electric field in the interpoly oxide, so finally the injection current stops its growth and starts to decrease. When the experiment ends, the injection current falls immediately down to zero. The average injection current from simulations was calculated with fairly good agreement with those calculated from experimentation.

The main result from this study is that under the conditions tested, the current necessary to perform charge injection in the floating gate is rather low, in the order of the tens of femptoampere for an output voltage applied to the injection electrode of approximately 25 V, which means that the voltage across the interpoly oxide is around 20 V. Considering the parasitic resistance in the experimental setup calculated in approximately 470 G Ω , in order to reach the injection voltage, it is necessary to force a constant current to the electrode of tens of picoampere, three order of magnitude higher. This means that the constant current strategy for charge injection it is not optimal in this context.

Nevertheless, this study was necessary in order to safely

characterize the voltage-current behavior of the injection process. Now that the injection dynamics are known, a better strategy can be implemented. We are now designing a close-loop injection circuit which implements a slow slew-rate voltage ramp and a current comparator for the drain current, which is the variable we want to control. We find that as the injection current is so low, a voltage controlled strategy would be better.

On the other hand, if a current-controlled strategy is desired, the area of the injection electrode should be much larger, but this strategy has several drawbacks. For starters, a larger area in the injection electrode impoverishes the capacitive coupling with the floating gate, as the interpoly capacitance will also be larger. Moreover, the current needed to mask the leakage current is 10 thousand times larger, so the injection electrode should increase its side 100 times, meaning that $A_{inj} = 600 \mu\text{m} \times 600 \mu\text{m}$, much larger than the floating gate area. Considering that the FG is augmented in the same factor, the transistor would not fit the die. And finally, if the current is 10000 times larger, the injection time will be 10000 shorter, and it may be difficult to control.

In conclusion, this study was very important to determine the best strategy that should be implemented in order to charge or discharge the floating gate transistor. Future work will be aimed to implement an injection circuit for latter characterization of the device as a radiation sensor.

REFERENCES

- [1] *Preventions of Accidents to Patients undergoing Radiation Therapy*. ICRP Publication 86. Ann ICRP 30 (3), 2000.
- [2] *Comprehensive QA for radiation oncology*. AAPM Report No 46, 1994
- [3] *Diode in vivo dosimetry for patients receiving external beam Radiation Therapy*. Report of task group 62 of the Radiation Therapy Committee, AAPM 2005
- [4] A.B. Rosenfed, "Electronic dosimetry in radiation therapy," *Radiation Measurements*, vol 41, pp. 134–153, 2006.
- [5] J. Lipovetzky, M.A. Garcia-Inza, S. Carbonetto, M.J. Carra, E. Redin, L. Sambuco Salomone, A. Faigon, "Field Oxide n-channel MOS Dosimeters Fabricated in CMOS Processes," *IEEE Transactions on Nuclear Science*, vol. 60, no. 6, pp. 4683–4691, 2013.
- [6] M. Garcia-Inza, M. Cassani, S. Carbonetto, M. Casal, E. Redin, A. Faigon, "6MV LINAC characterization of a MOSFET dosimeter fabricated in a CMOS process," *Radiation Measurements*, vol. 117, pp. 63–69, 2018.
- [7] M. Garcia-Inza, S. Carbonetto, J. Lipovetzky, M.J. Carra, L. Sambuco Salomone, E. Redin, A. Faigon, "Switched Bias Differential MOSFET Dosimeter," *IEEE Transactions on Nuclear Science*, vol. 61, no. 3, pp. 1407–1413, 2014.
- [8] S. Carbonetto, M. Garcia-Inza, J. Lipovetzky, M.J. Carra, E. Redin, L. Sambuco Salomone, A. Faigon, "CMOS Differential and Amplified Dosimeter with Field Oxide N-Channel MOSFETs," *IEEE Transactions on Nuclear Science*, vol. 61, no. 6, pp. 3466–3471, 2014.
- [9] M. Garcia-Inza, S. H. Carbonetto, J. Lipovetzky and A. Faigon, "Radiation Sensor Based on MOSFETs Mismatch Amplification for Radiotherapy Applications," *IEEE Transactions on Nuclear Science*, vol. 63, no. 3, pp. 1784–1789, 2016.
- [10] M. G. Inza, J. Lipovetzky, E. G. Redin, S. Carbonetto and A. Faigon, "Floating Gate PMOS Dosimeters Under Bias Controlled Cycled Measurement," *IEEE Transactions on Nuclear Science*, vol. 58, no. 3, pp. 808–812, 2011.
- [11] Y. Wang, N.G. Tarr, K. Shortt, I. Thomson, "A sensitive floating gate MOSFET gamma ray dosimeter," *Proceedings of IEEE Sensors 2003* (IEEE Cat. No.03CH37498), vol.2, pp. 1271–1276, 2003.
- [12] N.G. Tarr, K. Shortt, Y. Wang, I. Thomson, "A sensitive, temperature-compensated, zero-bias floating gate MOSFET dosimeter," *IEEE Transactions on Nuclear Science*, vol. 51, no.3, pp. 1277–1282, 2004.
- [13] J. Cesari, D. Gomez, M. Roca, E. Isern, A. Pineda, E. Garcia-Moreno, "Floating Gate P-MOS Radiation Sensor Charging Cycles Characterization," *IEEE Radiation Effects Data Workshop (REDW)*, 2014.
- [14] M. Brucoli, S. Danzeca, M. Brugger, A. Masi, A. Pineda, J. Cesari, et. al, "Floating Gate Dosimeter Suitability for Accelerator-Like Environments," *IEEE Transactions on Nuclear Science*, vol. 64, no. 8, pp. 2054–2060, 2017.
- [15] P. Pavan, L. Larcher, and Andrea Marmiroli. *Floating Gate Devices: Operation and Compact Modeling*. Springer Science & Business Media, 2007.
- [16] Lenzlinger, M., and E. H. Snow. "Fowler Nordheim Tunneling into Thermally Grown SiO₂." *Journal of Applied physics*, vol. 40, no. 1, pp. 278–283, 1969.
- [17] Oldham, Timothy R., and F. B. McLean. "Total ionizing dose effects in MOS oxides and devices." *IEEE Transactions on Nuclear Science*, vol. 50, no. 3, pp. 483–499, 2003.
- [18] Y. Nissan-Cohen and J. Shappir and D. Frohman-Bentchkowsky, "Measurement of Fowler-Nordheim tunneling currents in MOS structures under charge trapping conditions." *Solid-State Electronics*, vol. 28, no. 7, pp. 717–720, 1985.